

Figure 1: Deltacloud API entities

- Security groups
- Deltacloud works with 11 different compute APIs such as EC2, vSphere, etc, and five different storage APIs such as S3, Eucalyptus Walrus, Google Storage, etc.

Benefits of the Deltacloud API

The benefits include:

- Flexible migration from one cloud service provider to another
- Freedom in choosing a provider
- Quality of service

How Deltacloud works

By using the Deltacloud API, consumers need not interact with specific cloud providers. They can use a single API to interact with them all. In a nutshell, Deltacloud can be used as a facade for scripts to provision virtual machines in different cloud environments. It is implemented in the Ruby programming language specifically, using the Ruby framework, Sinatra. Deltacloud uses the notion of a driver specific to a cloud service provider, which is an adapter that can be used to connect to different cloud environments. All data is communicated via the REST interface to a Deltacloud server, which has a REST interface. The Deltacloud project provides a CLI (command-line interface) tool, as well as client libraries in Java, Ruby, C and Python.

The Deltacloud API provides the essential operations that consumers want from any management tool or application, for interacting with cloud services. Deltacloud is a server that acknowledges a standard REST API call for creating new instances on the compute resources, getting the instance's virtual hardware profiles, rebooting instances and image details, etc. Deltacloud also provides an API for creating buckets, reading or writing data, and manipulating the buckets and storage blobs.

The Deltacloud server translates requests into the appropriate API calls for the respective IaaS providers it supports. If a cloud service does not have a native API to perform specific operations, Deltacloud notifies that so that developers can work around it. Its website is

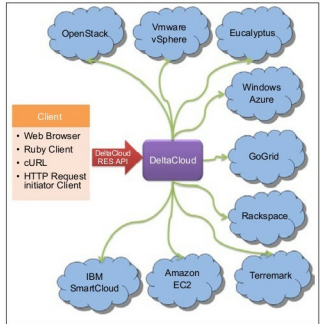


Figure 2: The Deltacloud API - How it works

deltacloud.org

How to install and run Deltacloud

Download Deltacloud from <http://deltacloud.apache.org/download.html>. Apache Deltacloud version 1.1.3 is available.

To install Deltacloud, you need to install its dependencies first: Ruby 1.8.7+, RubyGems 1.3.7+, G++, libxml2 and libxml2-dev, libssl and libssl-dev, and several Ruby gems, including thin, sinatra, rack-accept, rest-client, sinatra-content-for, and nokogiri.

The Advanced Package Manager (APT) commands (apt-get) take care of handling any other dependencies that are required.

To install the prerequisites on Ubuntu Linux, execute the following commands:

```
$ sudo apt-get update
$ sudo apt-get install ruby1.9.1-full
$ sudo apt-get install g++ libxml libxml2-dev libssl
libssl-dev
$ gem install thin sinatra rack-accept rest-client sinatra-
content-for nokogiri
```

Once all dependencies are installed, to install Deltacloud, execute the following command:

```
$ gem install deltacloud-core
```

Once installation is successfully completed, execute the following command:

```
$ deltacloud -i ec2 -r 192.168.1.2 -p 9988
```